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Protection of 100% Inverter-dominated Power Systems with Grid-Forming Inverters and Protection Relays – Gap Analysis and Expert Interviews

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ABSTRACT

Protection schemes for today's power systems have been developed over more than 50 years. These protection schemes assume that the power system and especially fault currents are dominated by synchronous generators in conventional power plants. However, the fast increase of renewable generation like wind and solar leads to power systems dominated by power electronic inverters instead of synchronous generators, e.g., in Hawaii and Texas. Synchronous generators and power electronic inverters show fundamentally different dynamics, especially during short-circuits. This may render today's protection schemes inadequate for these future renewable power systems.

This report summarizes a gap analysis resulting from a literature review and expert interviews conducted by subject matter experts from Sandia National Laboratory, Siemens, and the Electric Power Research Institute (EPRI) in Spring 2023. The gap analysis consists of two main parts: The fault-ride through (FRT) behavior of grid-forming (GFM) inverter-based resources (IBR) and the response of state-of-the-art protection relays to the fault currents and voltages from GFM IBRs.

Experts agree that GFM IBR dominated systems will lead to significant impact on system protection schemes, particularly distance, directional, and power swing schemes. Expert interviews and the literature review also confirm that a big gap exists between where the literature is today and what is needed for protecting a 100% IBR power system. There is hardly any literature or knowledge on

- a) measurements of fault currents generated by GFM IBRs,
- b) FRT strategies for GFM IBRs with focus on correct operation of protection relays,
- c) protection relay functions that respond correctly to fault currents measured in IBR-dominated systems,
- d) interaction between GFM and GFL IBRs during and after faults and their impact on the protection system,
- e) accurate models for GFM IBRs and protection relays for system protection studies,
- f) tools for protection studies in IBR-dominated systems, and
- g) input for future protection standards.

The experts agree that there is extensive R&D needed to fully comprehend the effect of GFM IBRs with different control structures and current limiting techniques for different protection functions during and after faults. This will then help in developing relay algorithms and protection schemes compatible with these sources.

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ACRONYMS AND TERMS

Acronym/Term	Definition
C&P	Control and protection
FRT	Fault ride-through
GFM	Grid forming
IBR	Inverter-based resources
PLL	Phase-locked loop
VRT	Voltage ride-through
WTG	Wind turbine generator

1. INTRODUCTION

Protection schemes for today's power systems have been developed over more than 50 years. These protection schemes assume that the power system and especially fault currents are dominated by synchronous generators in conventional power plants. However, the fast increase of renewable generation like wind and solar leads to power systems dominated by power electronic inverters instead of synchronous generators, e.g., in Hawaii and Texas. Synchronous generators and power electronic inverters show fundamentally different dynamics, especially during short-circuits. This may render today's protection schemes inadequate for these future renewable power systems.

Most inverter-based resources (IBRs) today are grid-following (GFL) and their behavior during faults – called fault-ride through (FRT) – is specified in standards like IEEE 2800. Power systems dominated by IBRs have to contain a significant portion of grid-forming (GFM) IBRs. GFM IBRs are substantially different than GFL IBRs, especially during faults. Today, there is very little knowledge about how GFM IBRs behave during faults and how protection relays respond to this behavior.

This report summarizes a gap analysis resulting from a literature review and expert interviews conducted by subject matter experts from Sandia National Laboratory, Siemens, and the Electric Power Research Institute (EPRI) in Spring 2023. This work is part of the Advanced Research Projects Agency-Energy (ARPA-E), U.S. Department of Energy funded project *PICo-Design: Protection-Inverter Co-Design for 100% Renewable Power Systems*.

The gap analysis consists of two main parts:

1. The fault-ride through (FRT) behavior of grid-forming (GFM) inverter-based resources (IBR) during under-voltage and fault events in Section 1 and
2. The response of state-of-the-art protection relays to the fault currents and voltages from GFM IBRs in Section 1.

Each section provides a summary of the literature review as well as the expert interviews. Section 2.3 also summarizes measurement results from GFM IBRs.

2. C&P FUNCTION GAP ANALYSIS FOR IBRS

The increasing penetration of IBRs is substantially reducing the share of synchronous generators (SGs) in power grids around the world. Thus, a part of the IBR fleet must take a direct role in regulating the voltage and frequency to maintain the system stability. The IBRs that offer such functionality are referred to as grid-forming (GFM) IBRs. Per recent grid codes and standards, during faults, GFM IBRs must remain connected to the grid and provide voltage support while complying with the fault ride-through (FRT) requirements of emerging regulatory frameworks, such as IEEE 2800–2022 standard [18]. The FRT capability of the GFM IBRs and their fault current signature directly impacts the operation of the protection system. This section summarizes existing works identified in the literature on different aspects of GFM FRT behavior.

2.1.1. FRT Standards for GFM IBRs

State-of-the-art grid codes and IBR interconnection requirements such as IEEE 2800-2022 specify IBR voltage ride-through (VRT) performance requirements that apply under abnormal voltage conditions including a fault (also referred to as FRT). The IEEE 2800-2022 does not distinguish between GFL and GFM IBRs, so the same requirements apply to both. The default reference point of applicability (RPA) for VRT requirements is the point of measurement (POM) which is on the high side of the main IBR transformer.

The VRT capability is expressed in terms of a voltage-versus-time curve which requires that for a given voltage, IBR plant shall not trip until the duration at this voltage exceeds ride-through curve capability. Figure 1 presents a sample voltage ride-through capability curve from IEEE Std. 2800-2022 [18]. The curve defines four operation modes, namely Mandatory Operation, Continuous Operation for 30 minutes, Continuous Operation for infinite time, and Permissive Operation. Mandatory Operation mode means that the IBR *shall* continue exchange of current between IBR and transmission system. By contrast, Permissive Operation mode means that the IBR is not required to continue exchange of current with transmission system; in this mode, the IBR *may* operate in either Mandatory Operation or current blocking mode which entails temporary blocking of controlled exchange of current.

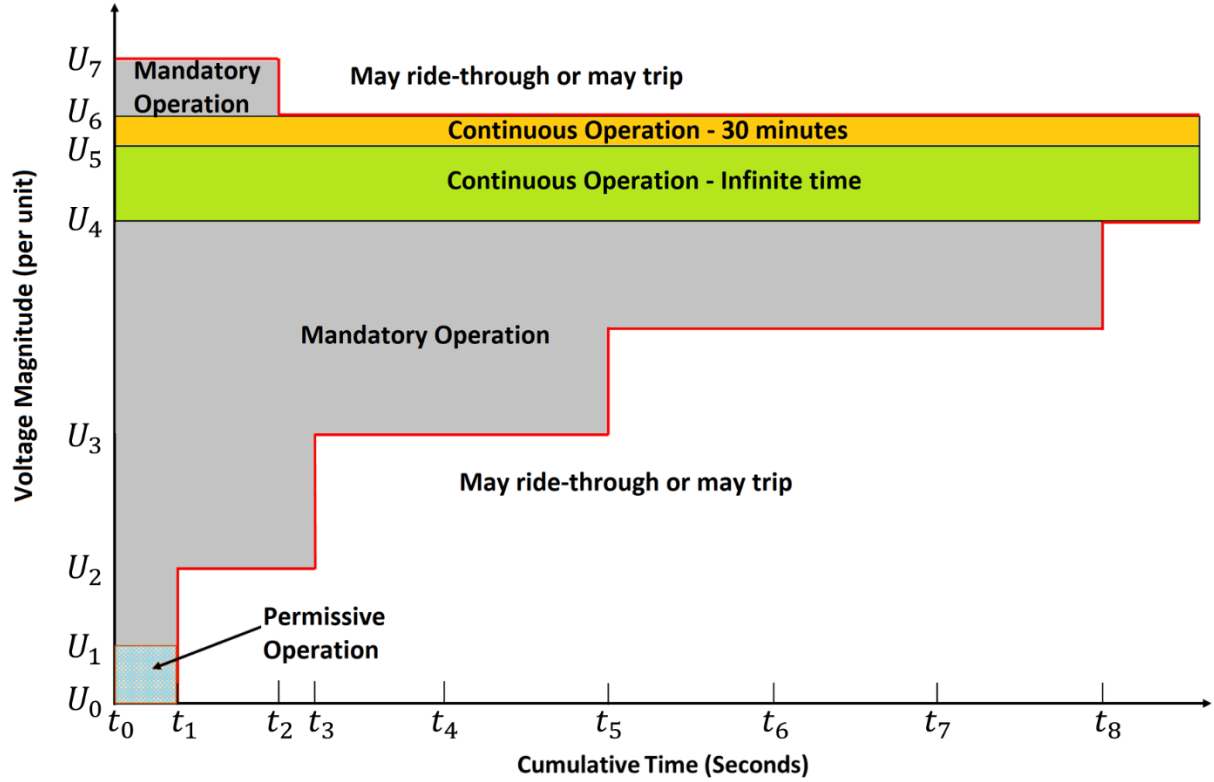


Figure 1: Sample Voltage Ride-Through Capability Curve based on IEEE Std. 2800-2022 [18].

IEEE 2800-2022 provides FRT requirements under both balanced and unbalanced faults as follows. Under a balanced fault:

- An IBR unit shall inject reactive current dependent on IBR unit terminal (POC) voltage;
- Under reactive/active current priority mode, the reactive/active current output is given priority and has the full current rating of the IBR available to it, while the active/reactive current output is constrained; and
- The difference between reactive current injection during a fault and a pre-fault reactive current output is an incremental positive-sequence reactive current. The incremental positive-sequence reactive current shall not be negative.

Under an unbalanced fault:

- In addition to increased positive sequence reactive current, the IBR unit shall inject incremental negative sequence current dependent on IBR unit terminal (POC) negative sequence voltage, and that leads the IBR unit terminal (POC) negative sequence voltage phasor by 90° - 100° for full converter-based IBR units and 90° - 150° for type III wind turbine generators (WTGs);
- In case IBR unit's total current limit is reached, either the incremental positive-sequence reactive current, or incremental negative sequence reactive current, or both may be reduced with a preference of equal reduction in both currents; and
- The incremental positive-sequence reactive current shall not be reduced below incremental negative sequence reactive current.

The standard does not specify the magnitude of injected incremental positive/negative sequence currents. The required step response time is not specified for type III WTGs and is less than or equal to 2.5 cycles for all other IBR units. The settling time is less than or equal to 6 cycles for type III WTGs and less than or equal to 4 cycles for all other IBRs. The settling band is a maximum of $\pm 10\%$ of required change or $\pm 2.5\%$ of IBR unit maximum current for all IBR units.

The standard further specifies requirements for restoring output after VRT. The following is a summary:

- Upon applicable voltage returning to continuous operation region, the IBR plant shall have capability to restore active power output to 100% of pre-disturbance level at an average rate equal to 100% of IBR continuous rating divided by specified active power recovery time;
- The default active power recovery time shall be 1s;
- The active power recovery time shall be configurable between 1.0 s and 10.0s;
- In weak grids, in order to reduce oscillatory behavior, it may be desirable to use slower active power recovery time;
- The active power recovery time shall be a target time and not be interpreted as a maximum time; and

Upon applicable voltage returning to continuous operation region, IBR units shall have capability to cease injection of incremental positive-sequence and negative-sequence reactive currents, per specified performance criteria.

2.1.2. **FRT Control Strategies for GFM IBRs**

The fault response of an IBR is largely dependent on the IBR control scheme and the FRT strategy it follows. There are two general strategies regarding the FRT of GFM IBRs in the existing literature,

1. switching to the grid-following (GFL) control structure upon detection of a fault and so operate as a current source [1] – [3].
2. maintaining the GFM control structure with a certain current limiting strategy [4] – [14].

In the first strategy, the IBR control system is switched to GFL via a fault detection algorithm that generates a trip signal in a short circuit condition. This method requires a backup phase-locked-loop (PLL) to synchronize to the grid, which results in losing the GFM operation during a fault. Also, it is necessary to return to the primary GFM mode by a reset signal when the short circuit fault is removed.

The second strategy, on the other hand, keeps the GFM control structure active during a fault and uses a current limiting technique to prevent an overcurrent. Different forms of current limiting techniques are proposed in the literature and will be discussed next.

Among these two strategies, the second one is largely popular in the literature mainly due to its capability to maintain the voltage source behavior of GFM IBRs during FRT. Hence, the following subsections review the existing literature on this category only.

Figure 2 illustrates the control system of a GFM inverter with the most common current limiting strategies [15]. The control system is composed of two control layers, i.e., the outer-loop control layer and the inner-loop one. The main objective of the outer-loop control is to synchronize the GFM inverter with the power grid and regulate the terminal voltage magnitude. A voltage magnitude E and phase angle θ will be generated based on the active and reactive power references P_{ref} , Q_{ref} , voltage, and angular frequency references, V_{ref} , ω_{ref} , and the feedback variables i , i_g , v_t , v_{PCC} . The inner

loop control aims to produce the voltage modulation reference v_{PWM} from V_{ref} and θ generated by the outer-loop control.

Figure 2. Control diagram of a GFM inverter with current limiting methods [15]

- ***Current magnitude limitation.*** The magnitude of the sustained GFM inverter's phase current should not exceed its maximum allowed limit, e.g., 1.2–2 p.u., to protect the power semiconductor switches.
- ***Fault current contribution.*** The GFM inverter should supply required active/reactive currents to support the power grid during disturbances.
- ***Fault recovery capability.*** The GFM inverter should be able to restore its normal operation when the fault is cleared.

- **Current limiter [4] – [7], [20] – [33]:** In this technique, the inverter control system restricts the phase current magnitude within the maximum allowed value through closed-loop current control [5] – [7]. This is often achieved by saturating the PI controllers of the inner current control loop and bypassing the outer loops. Consequently, appropriate techniques for avoiding the classical phenomena of “wind-up” or “latch up” should be adopted [4].
- **Virtual impedance [8] – [10], [34] – [45]:** This method adjusts the inverter’s phase current magnitude by increasing the total impedance based on virtual impedance/admittance control methods [8] – [10].
- **Voltage limiter [11], [12], [46] – [48]:** GFM control regulates the phase current magnitude by reducing the difference between the inner voltage and the point of coupling (POC) voltage via voltage limiters [11], [12].

- **Hybrid methods [13], [14], [28], [37], [43]:** This technique uses a combination of the above-mentioned methods to limit the inverter's current during FRT [13], [14].

The current limiters usually make the inverter behave as a current source during overcurrent conditions, which can facilitate the regulation of the output current vector angle to meet the fault current contribution requirement of grid codes. This mode is different from “switching to GFL” in the sense that GFM controls are maintained under a fault, and only current outputs are saturated. In comparison, the virtual impedance methods and voltage limiters can maintain the voltage source behavior of the GFM inverter to some extent during severe disturbances, which may allow for automatic fault recovery. The following subsections briefly describe the current limiting methods:

2.1.2.1. Current Limiter

In this current limiting mode, the inverter's current is limited directly by saturating the PI controllers of the inner current control loop. Based on how the saturated current reference is calculated, three current limiters are commonly utilized for GFM inverters, including the instantaneous limiter [20] – [22], the magnitude limiter [2], [23], [24], and the priority-based limiter [1], [25], [26]. Figure 3 illustrates different current limiters. In this figure, i_{ref} is the reference current from the outer loop, $\bar{i}_{ref}$ is the limited reference current, and I_M is the maximum current limit.

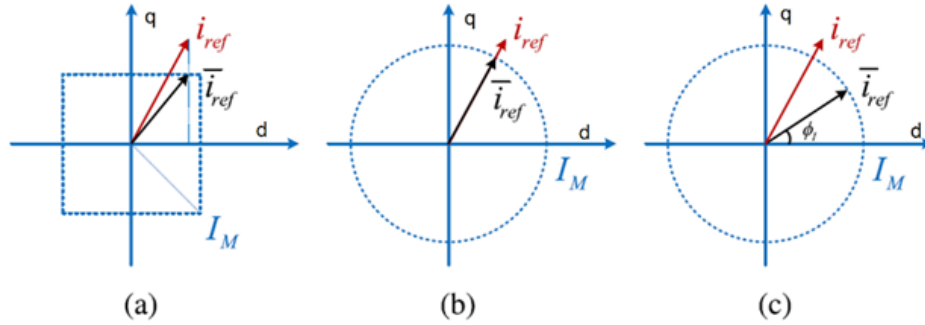


Figure 3. Illustration of different current limiters: (a) instantaneous limiter; (b) magnitude limiter; (c) priority-based limiter [15]

As shown in Figure 3(a), the instantaneous current limiter applies an element-wise saturation function to the instantaneous value of the reference current to achieve a limited current. When the current control is implemented in dq frame, there are two PI controllers, one for each I_d/I_q , and the instantaneous current limiter applies the limit to the d- and q-axis components of output current. When the control is implemented in alpha-beta frame, there are two proportional resonant (PR) controllers, one for each I_{α}/I_{β} , and the current limit is applied to the instantaneous alpha-beta-current components which are sinusoidal. In the abc-phase frame, there are three PR controllers, one for each $I_a/I_b/I_c$, and the current limit is applied to the instantaneous a-, b-, and c-phase current components which are sinusoidal. Note that Figure 3(a) shows the limiter for the dq frame implementation. On the other hand, the magnitude limiter in Figure 3(b) linearly scales down the magnitude of i_{ref} to I_M . Finally, Figure 3(c) shows the principle of the priority-based limiter, which not only decreases the magnitude of i_{ref} but also prioritizes its angle to a specific value φ_I . Note that φ_I is a user-defined angle that represents the angle difference between i_{ref} and the d-axis oriented to θ . This angle can be used for prioritizing active or reactive current output, e.g., as required by IEEE 2800-2022 Std. These current limiters can be adopted in different frames including dq , $\alpha\beta$, or abc frames.

All these three current limiters can achieve satisfactory current magnitude limitation under severe disturbances [15]. However, they can suffer from transient overcurrent caused by the current control loop dynamics. Among the three current limiters, the instantaneous limiter is the simplest one to achieve the overcurrent protection of GFM inverters. However, when it is implemented in the abc -frame and $\alpha\beta$ -frame, the inverter output current can become non-sinusoidal due to the clipping of the current reference [23]. Furthermore, in the $\alpha\beta$ -frame and dq -frame, conservative current limits may be needed, which can reduce the capacity utilization of the GFM inverter and requires the use of inverters with a relatively large maximum current limit, e.g., 2 p.u. in [22]. Compared with the instantaneous limiter, the magnitude limiter and the priority-based one can ensure a sinusoidal output current with extra current magnitude calculations, and fully utilize the overcurrent capability of the inverter during severe symmetrical disturbances.

In addition to the current magnitude limitation, these current limiters generally have difficulties in meeting the fault current contribution objective required by grid codes. When the current limiter is triggered, the inverter terminal voltage may not be aligned with the d -axis oriented to θ . This causes the phase angle to drift, and if another voltage source is present with a different frequency, such as a generator whose rotor speed is transiently dipping following a load step, the frequencies may diverge causing the dq voltages to “wind up.” Thus, the converter’s efforts to inject pure reactive current for voltage recovery would result in both active and reactive power exchange with the grid. To deal with this issue, one method is to switch the GFM inverter to a PLL-synchronized GFL inverter. However, the PLL can suffer from small-signal or transient instability issues under weak grid conditions [5], [27]. An alternative solution that avoids switching the synchronization methods is to adjust the active and reactive power references based on grid codes [28], [29], which are further tracked by power control loops to achieve the fault current contribution goal.

Furthermore, once these current limiters are triggered, the inverter phase current magnitude will be fixed at I_M . Therefore, the voltage vector controller and the voltage magnitude controller cannot adjust the inverter terminal voltage magnitude to its reference value. The windup of these voltage controllers may further result in a current reference whose magnitude is larger than I_M . Consequently, undesired current saturation occurs, which prevents the inverter from successful fault recovery [30]. Extra anti-windup designs are required for these voltage controllers to address this issue when current limiters are utilized. Moreover, it has been reported in the literature that using current limiters can jeopardize the stability of the inverter control system and excite the resonance frequency of the RLC filter [31] – [33].

2.1.2.2. Virtual Impedance

A widely used technique in the literature for limiting the GFM inverter’s current is the virtual impedance method. The main idea of this method is to add a large virtual impedance between the converter and the grid in the event of an overcurrent and so limit the current in this way [34]. The proposed scheme of [34] is activated upon detection of an overvoltage and is inconsequential under normal operating conditions. The scheme uses a PLL, and the turning on/off of the virtual impedance does not abruptly change the reference voltage angle. As shown in Figure 2, the voltage drop on the virtual impedance is subtracted from the reference voltage E generated by the outer control loop. As a result, the reference for the inner current control loop is reduced and so the inverter’s current is curtailed.

Unlike the current limiter methods, the virtual impedance method does not result in wind up of the outer control loops and provides automatic fault recovery. It also maintains the voltage source

behavior of the GFM during most fault conditions. However, due to the limited bandwidth of the voltage control loop, this method cannot limit the inverter's current immediately, especially during the first cycles of a fault [33]. Although this is a critical aspect for a current limiter, it has been neglected in many of the existing literature that develop FRT strategies with virtual impedance [14], [35] – [37].

Three types of virtual impedance methods have been used in the literature, namely virtual impedance with inner-loop control [9], [31], [38], virtual impedance without inner-loop control [10], [39], and virtual admittance [4], [40]. In the implementation of virtual impedance with inner-loop control, it is assumed that the voltage after the RLC filter is equal to the reference voltage from the outer loop. Therefore, the RLC filter impedance is also considered in the virtual impedance value, and so the voltage modulation reference is modified directly. An equivalent circuit diagram of this method is demonstrated in Figure 4(a). In the virtual impedance without inner-loop control implementation, the RLC filter impedance is not considered in the virtual impedance value, and these two impedances are in series. The circuit diagram of this method is shown in Figure 4(b). Compared to these two implementations that require a derivative controller to achieve a virtual inductor or a virtual reactor at the nominal frequency, the virtual admittance method can achieve a virtual inductor within the bandwidth of the current control loop. However, the virtual admittance method cannot have zero resistance and inductor in normal operation, whose selection may not follow the procedure of the two above-mentioned implementations. The circuit diagram of this method is depicted in Figure 4(c). The difference between the control of Figure 4(a) and Figure 4(c) is that the former provides a virtual reactance X_v at the nominal frequency, whereas latter can realize a virtual inductance L_v within the bandwidth of the current control loop.

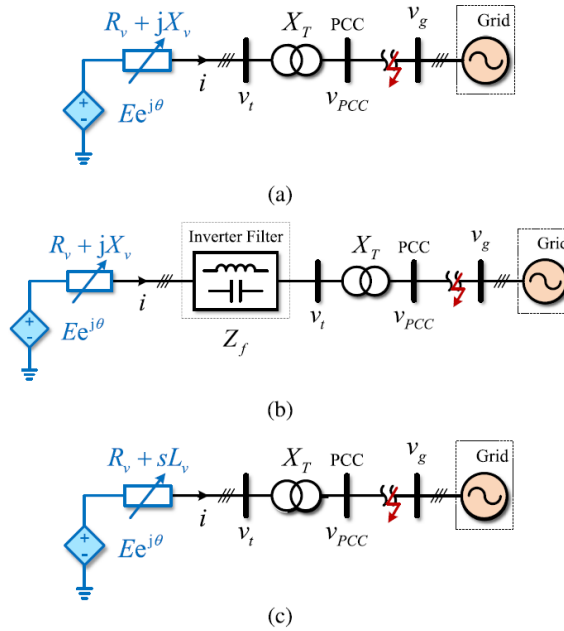


Figure 4. Equivalent circuit diagram of different virtual impedance control methods. (a) Virtual impedance with inner-loop control. (b) Virtual impedance without inner-loop control. (c) Virtual admittance [15].

The virtual impedance without the inner-loop control and the virtual admittance method with a fast-tracking current control loop can achieve good current limitation performance when severe disturbances occur [41]. In comparison, the virtual impedance method with inner-loop control

achieves current limitation based on the hypothesis that the outer-loop voltage reference can be tracked quickly by the voltage control loop. Since the bandwidth of the voltage control loop is typically low, temporary overcurrent may be observed [42], [43]. To deal with this issue, hybrid current limiting methods that combine the virtual impedance with the current saturation method are presented [13], [14].

The magnitude and X/R ratio of the virtual impedance can be selected to be constant during FRT or adaptively change depending on the system condition. Existing virtual impedance models are typically designed to have a constant impedance and limit the inverter's current to a constant threshold, i.e., the inverter's maximum current limit [31], [33]. Although this can result in limiting the inverter's current for symmetric faults, it fails to achieve this objective for asymmetric faults [13]. Moreover, this approach works within a certain range of grid impedances, and its performance can be adversely impacted in case of grid impedance variations and different voltage dip levels [44]. Conversely, the adaptive virtual impedance method can achieve current limitation during all types of faults [4], [8], [13], [45]. However, it is important to appropriately select the magnitude and angle of the virtual impedance to achieve a perfect balance between stability, current limitation, and voltage support. It should be mentioned that under this method, the virtual impedance is adaptive to terminal voltage v_t and current i , and hence may change during a fault. As such, the impedance is also implicitly adaptive to the type and location of fault. A small X/R ratio, i.e., a more resistive virtual impedance, results in a less oscillatory response with a lower possibility of overcurrent, but it also makes the GFM's current mostly active, thereby compromising voltage support during faults. Additionally, a virtual impedance with a large magnitude keeps the inverter's current below the limit but increases the risk of instability. It is also worth mentioning that since the voltage source behavior of the GFM is maintained if the virtual impedance method is used, the angle of the inverter's sequence currents during FRT depends on the magnitude and angle of the virtual impedance and the physics of the system, such as the fault resistance and line impedance [13].

2.1.2.3. Voltage Limiter

This current limiting strategy directly reduces the voltage difference between the modulated voltage, v_{PWM} , and the terminal voltage, v_t , in Figure 2 to smaller than a certain threshold [11], [12], [46], [47]. This method is a suggested solution in [16] since it does not require adaptive virtual impedance that can destabilize the system under certain conditions [48]. For the voltage limiters, the inner-control loop is commonly transparent, i.e., $v_{PWM} = v_{ref}$. Subsequently, an equivalent circuit diagram of this current-limiting method can be expressed as in Figure 5.

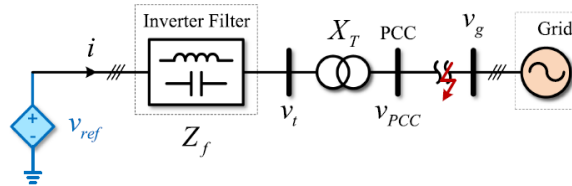


Figure 5. Equivalent circuit diagram of voltage limiter method [15]

The implementation of the voltage limiter is usually achieved by regulating E and θ generated by the outer-loop control where these quantities are evaluated to be within certain thresholds to ensure that $v_{ref} - v_t$ does not result in overcurrent. In [46] and [47], a voltage limiter is directly designed in the abc frame where the voltage limits are calculated based on each phase current. This type of voltage limiter is simple in implementation yet introduces non-sinusoidal phase currents like the instantaneous

current limiter. Besides, to ensure the fault recovery capability, appropriate anti-windup designs for outer-loop controllers are required due to the saturation of the control signals E and θ . When both E and θ are saturated, v_{ref} must follow the changes in v_t . Appropriate selection of control parameters, such as the thresholds, is necessary to meet the FRT current requirements of different grid codes and standards. Moreover, similar to the virtual impedance method and current limiter, voltage limiter can also experience temporary overcurrent transients in the first few cycles of a fault.

2.1.2.4. Hybrid Methods

To overcome the shortcomings of different current limiting technique and, at the same time, take advantage of their benefits, various current limiting strategies can be combined to form a hybrid method [13], [14], [37], [28], [43]. As an example, [13] uses the current saturation method during transients and the virtual impedance method at steady state to limit the inverter's current and also maintain the voltage source behavior of the GFM.

Table 1 provides a comparison between different current limiting strategies.

Table 1. Performance comparison of different current limiting strategies [15].

	Current limitation performance	Fault current controllability	Fault recovery capability
Current limiter	• Good in transients and steady state • Potential non-sinusoidal current • Possible temporary overcurrent in transients for certain implementations	• Change to PLL-based GFL inverter and adjust the current references • Adjust power references	May fail to recover due to the windup of voltage controllers
Virtual impedance	• Trade-off between current limitation and stability • Potentially slow current limitation • Temporary overcurrent in transients	• Depends on the virtual impedance and electrical network parameters • Adjust power references	Good to restore normal operation automatically
Voltage limiter	• Good in the steady state • Potential non-sinusoidal phase current • Temporary overcurrent in transients	Depends on the voltage limiter and electrical network parameters	May fail to recover due to the windup of voltage controllers
Hybrid Methods	• Good in transients and steady state • Reduced likelihood of overcurrent in transients	Depends on the type of the combined current limiters	Depends on the type of the combined current limiters

2.1.3. *FRT Strategies that Meet Grid Codes*

Various FRT strategies for GFM IBRs are available in the literature. Yet, due to the lack of clear specifications regarding the behavior of GFM IBRs during faults, in most of the cases, the focus has been on limiting the inverter's current and preventing instability without maintaining the GFM behavior during FRT. In recent years, industry stakeholders have started working on developing grid codes and standards to specify performance requirements for GFM IBRs. Currently, there are ongoing efforts in Great Britain, Germany, across Europe (through ENTSO-E), Hawaii, and Australia [49]. Moreover, the requirements in the recently published IEEE 2800-2022 standard [18] are expected to apply to GFMs.

There are a few works in the literature that have developed FRT strategies that meet grid code requirements. The authors of [13] have proposed a control scheme that can make GFM IBRs comply with the negative-sequence current low-voltage ride-through requirements of the IEEE 2800–2022 standard. This scheme acts as a controllable reactive current source in the negative-sequence circuit, i.e., the IBR injects a purely reactive negative-sequence current during low-voltage events. Additionally, it maintains the voltage source behavior of the GFM in the positive-sequence circuit. The proposed method uses an adaptive approach to divide the inverter's current generation capacity between the positive- and negative-sequence currents so that the maximum capacity of the inverter is utilized during faults. This scheme uses a hybrid current limiting strategy by combining the virtual impedance and current limiter methods to prevent overcurrent during transients and steady state.

As another work, [4] proposes an FRT control system for GFM IBRs that meets the requirements of the British system operator NGESO [50]. Firstly, the paper highlights the requirements of this grid

code. The proposed control system complies with the above requirements by calculating a proper virtual electromotive force (EMF) and an adaptive virtual impedance. The currents are limited using two approaches: i) limiting phase currents directly ii) limiting positive- and negative-sequence currents. The currents generated through each of these approaches meet the grid code requirements but have different characteristics. The second approach results in a more balanced current during symmetrical and asymmetrical faults, which is not desirable for protection purposes. Moreover, although the proposed controller injects a highly reactive current in the positive-sequence circuit, the negative-sequence current is not necessarily inductive, as it is not regulated directly, which can jeopardize reliable operation of negative-sequence-based protection elements.

Another work on the FRT strategies that meet grid code requirements is [36]. This paper proposes an auxiliary control system for IBRs that is added to the main controller and enables ride-through capability of the inverter. The proposed controller receives the active and reactive power setpoints for the positive- and negative-sequence loops and generates proper references for the inner control loops. The setpoints can be generated based on a certain grid code. Compliance with Danish and German codes for IBRs is demonstrated in the paper. A voltage dip detection algorithm is also implemented to activate the proposed controller in the event of faults. Moreover, the overcurrent is avoided by adjusting the power setpoints. There is no clear discussion in the paper on whether the voltage source behavior of the GFM is maintained in the positive-sequence circuit, but since the active and reactive power setpoints are directly regulated, it can be said that the proposed method loses this functionality.

Finally, [51] proposes two FRT strategies for GFM IBRs that can meet grid codes. The paper firstly provides a discussion on how a direct compliance with the reactive current requirements of grid codes can be challenging for GFM IBRs due to not having a PLL to capture the grid voltage angle. It then proposes an FRT strategy to overcome the challenges. This method is similar to switching to a GFL structure with one difference that the voltage angle at the PCC is directly fed back to the control system instead of having a PLL. An anti-windup technique is then adopted for the outer control loops. Acknowledging the limitations of the first method, the paper proposes a second FRT method that acts directly on the voltage references generated from the outer loop, similar to the virtual impedance method, to achieve a performance level similar to what grid codes desire. Two approaches are taken that include voltage balancing priority (method A) and voltage magnitude priority (method B). The main purpose is to directly define the desired AC voltage profile during fault; in case a balanced voltage but with reduced magnitude is preferable, method A could be used; whereas, if an unbalanced voltage but with higher magnitude is needed, method B could be implemented. Although these methods can provide certain benefits from power system perspective, they do not meet the requirements of existing grid codes and also do not guarantee reliable operation of protection devices.

2.2. Expert Interview

We interviewed a total of 18 experts. The focus was on getting feedback from inverter/protection vendors, consultants, national laboratories, and utilities. Experts from the following organizations participated in the interviews: Electranix, Sandia National Laboratories, EPC Power, Siemens PTI, Siemens Gamesa, Southern Company, Quanta, ETAP, GE, Hickory Ledge, RTE France, Dominion Energy, and Hawaiian Electric. Of the 22 total expert interviews 5 were conducted only within Siemens with existing non-disclosure agreements (NDAs). Care has been taken to exclude information that said organizations are not willing to publicly share. The following section provides a summary of the key take aways from the interviews.

1. *What are known issues in real-world over-current/protection behavior in inverter-dominated systems, either microgrids, distribution systems, or transmission systems*

It was agreed among all experts that IBRs (not limited to GFM) have to provide negative sequence current during fault for protections to work accordingly. Many GFL IBRs installed today are still only providing positive sequence fault current. Inverter manufacturers are gradually adding this function to their inverters, this strengthens the importance of incorporating negative sequence current capability into GFM IBR. Zero sequence current is not a big problem for transmission system as delta-Y transformers usually help in providing it and mitigating the fault. Moreover, due to the physical constraints of inverters, IBRs cannot provide enough fault currents to trigger over-current-based protection devices. This is an open space for research and study. Some experts pointed out that in transmission systems multiple GFM IBRs may be able to provide high short-circuit currents collectively because they are not all operating at their current limit before the fault. Another practical issue for current protection schemes is that IBRs usually trips before protection devices due to the fastness of IBR response. This stresses the need for orchestration between GFM/GFL control and protection with system protection scheme. Regarding protections of GFM IBRs, the experts have not seen existing cases in which GFM IBR connected to transmission system trips, and it is also unclear how BESS GFM should behave during fault if it is in charge state.

2. *What is state-of-the-art grid-forming (GFM) inverters behaviors during over-current situations?*

Although GFM inverters from different vendors behave differently during saturation, experts report similar patterns when GFM inverters enter saturation:

1. the HW limit and fast opening of switches will lead to distorted wave forms in the first few cycles,
2. after that, the software control, e.g., current loop brings the current back to sinusoidal waveform.

During saturation, most GFM inverters provide at most 1.1-1.2 p.u. fault current.

3. *Expected GFM inverter behaviors during saturation, and new expectations other than fault ride-through in inverter-dominated systems*

The following functions and capabilities are identified by experts that are needed for GFM inverters during saturation.

- The GFM inverters should provide sinusoidal current and voltage during fault in order for traditional protection relays to work.
- HW-level blocking of transistor switching to protect inverter HW is acceptable for the first few cycle as long as it transitions into sinusoidal currents and voltages afterwards. GFM inverters are supposed to control the phase angle of the fault current during fault.
- GFM inverters should comply with IEEE 1547 and 2800 standards during faults, which are actually generic for both GFL and GFM IBRs. It is identified that existing GFM IBRs in the system usually struggle to comply with these standards.

4. *Should a GFM IBR turn into GFL mode during fault ride through?*

The importance of GFM IBRs during faults are well recognized among all experts in the sense that the voltage source properties of a GFM IBR is mostly needed during fault ride-through. Most experts in the interview (also working groups cited) agreed that mode switching from GFM into GFL or GFL FRT should be avoided. GFM IBRs should continue to act in GFM mode during faults to support voltage while limiting its current output under its operational range. One expert pointed out that switching from GFM to GFL IBR can provide better FRT and fault current, however, the mode switching is problematic during faults and is prone to oscillations and chattering between operation

modes of different IBRs. In cases where a GFM IBR has to change to GFL due to physical constraints when it cannot comply with full GFM characteristics, the mode switching from GFM to GFL should be smooth, i.e. the switching response should not be noticeable from outside.

5. Should GFM IBRs provide positive and negative sequence injections during fault?

As today's protection schemes are mostly dependent on negative sequence current, the interviewed experts agreed that GFM IBR (no different than GFL IBR or even synchronous machine) should provide negative sequence current during fault. One expert pointed out that it is also possible for GFM IBR to provide zero sequence current. However, this requires hardware additions to inverters and adds additional cost to inverter manufacturers. Therefore, it is agreed that it is still preferable to provide zero sequence current from transformers instead of from GFM IBRs. We have also identified a gap from experts on operational side, that the ratios between positive and negative sequence current should be tunable and controlled based on operating condition. We confirmed from one inverter manufacturer that such features are available in their inverter specs.

6. What should be the fault recovery capability of GFMs - able to restore its normal operation when disturbances are cleared?

The experts agreed that fault recovery capability is not so different between GFM and GFL IBRs. The minimum requirement is that GFM IBR should stay connected to the system and be stable after the fault clearance. More specifically, GFM IBR should maintain GFM mode as much as possible during fault. After fault, GFM IBR should recover the voltage after fault, then provide frequency support. The voltage and frequency recovery rate from GFM IBR should confirm to standards or system requirements to avoid out-of-step oscillation. A challenge is recognized regarding the control of phase angle and subsequently re-synchronization of GFM IBR to the grid. Moreover, in blackstart scenarios, GFM IBRs should be able to smooth out voltage spikes after fault due to load decrease until the load can be reconnected again.

7. What are the functions and requests related to protection vendors get from customers to include in the GFM inverter nowadays?

The experts we interviewed are not aware of specific requirements from customers regarding protections of GFM inverters. The only functions customers ask are related to variations of inverter function to follow grid codes with no distinction between GFL or GFM inverters.

2.3. Measurements

Measuring the response of GFM IBRs to faults is essential to ensure that they can provide reliable and safe operation in the event of grid disturbances. Experiments to measure the response of GFM IBRs to faults typically involve subjecting the inverter to various fault conditions and monitoring its response. These faults may include short-circuits, ground faults, overvoltage or undervoltage conditions, and other disturbances that can occur in the grid. The experiments are typically conducted in a controlled environment using laboratory equipment or simulation tools.

One way to conduct experiments is to use a power hardware-in-the-loop (PHIL) setup. This involves connecting the GFM IBRs to a real-time digital simulator that emulates the power system's behavior. The simulator emulates faults and disturbances in the simulated grid, and the inverter responds accordingly. The PHIL setup allows for accurate and repeatable testing of the inverter's performance in a variety of scenarios.

Another method to test GFM IBRs is to connect them directly to a controllable load bank and perform load steps significantly larger than the inverter size. For example, putting a 500-kW load on Phase A

of a 100 kVA GFM IBR, the response is similar to a single-line-to-ground fault. Due the complexity of synchronizing GFM IBR with a grid-simulator in PHIL, all test results below were achieved with this method of using large load banks.

In [52], two GFM IBR from different manufacturers were tested. Both were single-phase GFM inverters designed for split-phase 120/240V off-grid PV with energy storage applications. As seen in Figure 6, both inverters provided square-wave currents during faults, which increases the total magnitude of the current but also introduces harmonics. The figure shows the voltage and current waveform data as solid lines, and the calculated RMS values for voltage and current are shown as dashed lines. Note that a resistive load bank was used for this test, so only real power is injected with the current staying in phase with the voltage during the fault. On the left side of Figure 6, the fault was applied at 0.1 seconds, and the calculated RMS current goes to 2.25 pu during the fault. For a different inverter on the right side of Figure 6, the fault was applied at 0.5 seconds, and the calculated RMS current goes to 1.3 pu during the fault.

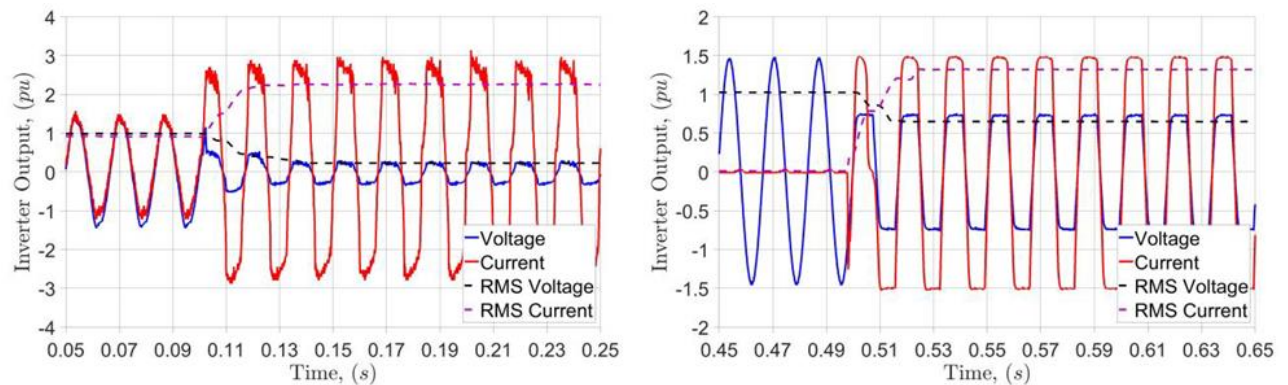


Figure 6. Experimental fault testing of single-phase GFM IBRs from two different manufacturers

In [53], two three-phase GFM IBRs were tested under fault conditions by applying large loads. Some example results can be seen below in Figure 7. The top three graphs show the voltage and current waveform data for each of the three phases, along with the calculated RMS values. The bottom plot shows the calculated sequence currents (positive, negative, and zero) noted as “Sym (I).” The first GFM IBR on the left is shown injecting positive, negative, and zero sequence square-wave currents during a single-line-to-ground fault tested by applying a 6pu load to Phase A. This inverter is composed of three individual single-phase inverters tied together through a control scheme to have a master device and two slaves. This configuration allows zero sequence current to be injected directly from the inverter without a transformer. Note that the positive, negative, and zero sequence currents are all equal, as would be expected for a single-line-to-ground fault. The second GFM IBR on the right side is shown injecting positive and negative sequence sinusoidal currents during a high-impedance line-to-line fault. This inverter cannot provide zero sequence current, but it is connected through a delta-wye transformer that provides the zero-sequence path. Note that the positive and negative sequence currents are equal, as would be expected for a line-to-line fault.

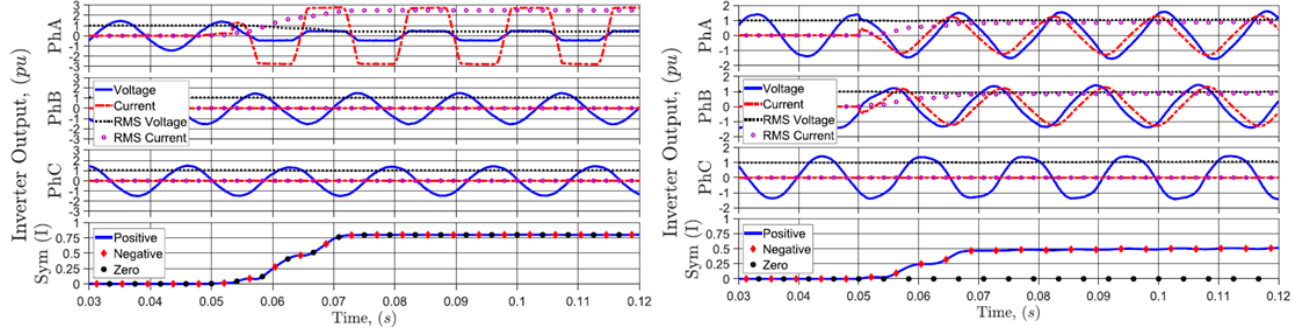


Figure 7. Experimental fault testing of 3-phase GFM IBRs from two different manufacturers.

Some of these same inverters were tested in [54], a single-phase and a three-phase GFM IBR that both provided square wave current outputs. While the fault tests are not necessarily new, the interesting aspect of this work was developing models that represented the same characteristics as the experimental tests. The models performed very similarly to the hardware experiments during faults. As seen in Figure 8, both device-under-test (DUT) 1 and 2 were both accurately captured by the model.

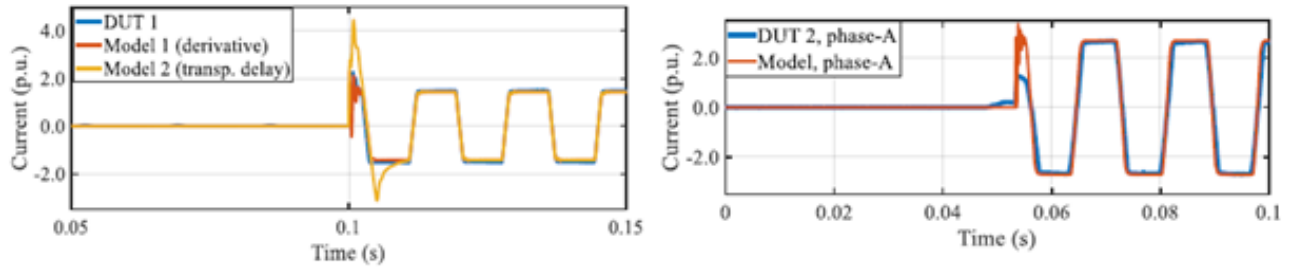


Figure 8. Experimental fault tests of a) single-phase GFM IBR, and b) 3-phase GFM IBR being used to validate simulation models of GFM IBR

Faults also generally create a sudden jump on the phase angle of the system voltage due to the change in impedance, or in the case of a line-to-line fault, the voltage jumping towards each other. In [55], using a grid-simulator, phase angle jumps were applied at the terminal of a 3-phase GFM IBR in different control modes. While the phase angle jump pulls large amounts of current from the GFM IBR, it stabilizes within 10 cycles. See the reference for the experimental results and figures.

3. C&P FUNCTION GAP ANALYSIS FOR PROTECTION DEVICES

3.1. Literature Review

Unlike GFL IBRs, the impact of GFM IBRs on the protection system has remained a largely unexplored area in the literature, except for the impact on microgrid protection [56,57]. It is critically important to deeply investigate how GFMs with different FRT strategies and current limitation techniques affect various protective devices in their vicinity.

The most relevant work to this topic is [58]. This work conducts a preliminary study on the impact of GFM IBRs on the distance relay that protects the GFM tie-line. Three types of GFM control structures are modeled;

- i. dq frame with only positive-sequence current control equipped with current limiter (M1),
- ii. $\alpha\beta$ frame control with current limiter (M2), and
- iii. $\alpha\beta$ frame control with virtual impedance (M3).

Unlike the M1 controller, M2 and M3 controllers allow the flow of an uncontrolled negative-sequence current during faults. The paper evaluates the performance of the distance relay during phase-B-to-Phase-C (BC) faults on different locations of the IBR tie-line when the GFM uses different controllers. The fault resistance value is not mentioned, and the relay uses apparent impedance to determine the fault zone. It is shown that when the GFM uses M1 and M2 controllers, the relay misoperates by measuring an impedance in the fourth quadrant for internal faults and an impedance in zone 1 for external faults, i.e., the relay over-reaches. However, for the GFM with M3 controller, the relay performs much better and measures an apparent impedance with a small error for most faults. The paper relates this superior performance to the fact that the M3 controller uses a virtual impedance with an X/R ratio that matches that of the line impedance.

Although this work can be a good starting point for investigating the impact of GFM IBRs on the protection system, it suffers from the following shortcomings:

- The work relies solely on the simulation results and does not provide any analysis or insight into the reasons behind the obtained results.
- The simulations are conducted for only one type of fault with an unknown fault resistance value.
- Some of the most recent control structures for GFMs, e.g., IEEE 2800-2022 compliant negative-sequence current injection, are not considered.
- A simplified implementation of distance relays is considered, i.e., apparent impedance.
- No sensitivity analysis on the impact of different control system and network configuration parameters on the performance of the relay is provided.

In [59], a very basic simulation analysis was performed for GFM IBR impacts on a 46 kV sub-transmission line protection schemes. Inverse time-over current, distance protection, and direction elements were studied. While no major difficulties were found for the protection schemes, the GFM IBR provided very distorted non-sinusoidal that may impact the algorithms in the relays.

Table 2 provides a summary of the protection-related works on GFL and GFM IBRs.

Table 2. Summary of literature on protection impacts by GFL and GFM IBRs

Protection scheme	Expected GFL impact	Expected GFM impact
Power swing protection	References [60] – [62]: • Potential misoperation of Power Swing Blocking due to reduced inertia and increased rate of change of swing impedance. • Potential Out-of-Step Tripping malfunction due to changed swing impedance trajectory. • Changed location of Electrical Center.	No literature found.
Negative-sequence-based protection	References [63] – [73]: • Negative-sequence overcurrent element may not pick up due to the low negative sequence current contribution of IBRs. • Directional negative-sequence overcurrent may malfunction due to the unconventional negative sequence current angle.	No literature found.
Phase selection element	References [64], [72]: Phase selection element may fail to correctly identify the faulty phase due to the unconventional sequence current angles of IBRs.	No literature found.
Communication assisted protection	References [64], [72]: Permissive Over-Reaching Transfer Trip (POTT), Permissive Under-Reaching Transfer Trip (PUTT), Directional Comparison Blocking (DCB), and Directional Comparison Unblocking (DCUB) may malfunction due to the failure of a supervising directional element.	No literature found.
Line distance protection	References [73] – [84]: • Low fault current amplitude due to IBRs may lead to lack of enough supervising current, thus leading to failure to trip. • Dynamically varying source impedance (and angle) of IBRs may cause unpredictable and inconsistent dynamic expansion of mho circle, thus leading to reduced reach accuracy and risk of over- or under-reach.	Reference [58,85]: • GFM IBRs with current limiter can cause under/overreach of a distance relay. • GFM IBRs with virtual impedance can potentially enable correct operation of a distance relay.
Memory-polarized zero-sequence directional element	Reference [86]: Faster IBR controls and reduced inertia may cause a shift in the phase angle of the short-circuit voltage with respect to that of the memory voltage, leading to an incorrect directionality decision.	No literature found.

Protection scheme	Expected GFL impact	Expected GFM impact
Line current differential (LCD) protection	Reference [87]: - Traditional slope-based LCD seems to be immune to the IBR fault behavior. - Alpha-plane LCD is prone to misoperation due to the lower fault current amplitude and varying current angle of IBRs.	No literature found.
Traveling wave protection	References [88]-[90]: - Inverter controls do not respond fast enough to have any impact on the traveling wave arrival times, magnitude of incremental quantities, etc. - IBR filter impedance and GSU parameters impact the reflections of the traveling wave	

3.1.1. Impact of IBR FRT Performance on System Protection

Recent grid codes and IBR interconnection standards such as IEEE 2800-2022 require specific IBR performance under a fault which can potentially impact the performance of system protection. Table 3 summarizes the identified impacts in the literature.

Table 3. IBR FRT impact on protection

IBR performance	Definition	Impacted protection
Voltage Ride-through	Under a low/high voltage condition, IBR plant is required to maintain its operation and stay connected to the grid for a specified duration.	IBR current injection during a fault could be potentially favorable or unfavorable from protection system perspective. The favorable impact includes grid voltage support during faults. Unfavorable impacts include potential interference with distance and overcurrent protection & driving fault currents beyond equipment limit.
Positive sequence reactive current support under FRT	Under FRT, IBR unit is required to inject positive sequence reactive current dependent on IBR unit terminal voltage.	Some impedance protection, overcurrent protection, directional overcurrent elements rely on positive sequence currents to operate reliably. More studies are needed to characterize the impacts.
Negative sequence reactive current support under FRT	For unbalanced faults, in addition to the positive sequence reactive current, IBR unit is required to also inject negative sequence reactive current.	- A number of impedance-based protection elements rely on negative sequence short-circuit current injection to operate reliably & selectively. - A number of overcurrent protection elements rely on the magnitude of negative sequence current to detect an unbalanced fault. - A number of directional overcurrent elements rely on the power factor angle of negative sequence current to make a directionality decision. - Fault identification: This element relies on the power factor angle of negative

IBR performance	Definition	Impacted protection
		sequence current to identify the faulted phase. • Communication-assisted protection: Some schemes use a directional element relying on the power factor angle of negative sequence current. • Other impacts: IBRs are current limited devices. Even when required to inject negative sequence current during unbalanced faults, the magnitude of this current would be limited. Further, increased negative sequence current injection could reduce the magnitude of positive sequence current injection which could impact the performance of other protection elements. As such, the full extent of IBR negative sequence current injection on traditional protection schemes is still unknown especially when the penetration of IBRs is significantly high throughout the interconnection.
Frequency Disturbance Ride-through	Under a frequency disturbance, IBR plant is required to provide active power requirements of primary frequency response and/or fast frequency response as applicable or maintain pre-disturbance active power output.	• Power swing detection and protection: IBR frequency response can impact the performance of power swing blocking function which relies on the rate of change of swing impedance trajectory which depends on system inertia. More studies are required to characterize the impact. • Rate-Of-Change-Of-Frequency (ROCOF) protection: IBR frequency response may also impact the performance of ROCOF protection. More studies are required to characterize the impact. • Under-Frequency Load Shedding (UFLS): IBR frequency response may impact ROCOF and frequency nadir following a frequency event, thus potentially impacting the performance of UFLS.

3.1.2. Standards for Protection of Inverter-based Systems

The growth in inverter-based generation has prompted the development of standards to protect these systems. The IEEE Power System Relaying and Control Committee (PSRC) has established several working groups for development of these standards, such as:

- C32 Protection Challenges and Practices for Interconnecting Inverter Based Resources to Utility Transmission Systems
- C45 Protection and short-circuit modeling of systems with high penetration of inverter-based resources

Despite the existence of these standards, there are still some gaps in the protection of inverter-based generation. One of the main gaps is the focus of existing standards and research around GFL IBR's

response to faults. There is a lack of standards for the integration of energy storage systems or inverters with GFM controls. As the use of renewable energy and GFM IBRs continues to grow, it is important that standards are in place to ensure the safety, reliability, and security of these systems.

3.2. Expert Interview

We interviewed a total of 22 experts. The focus was on getting feedback from inverter/protection vendors, consultants, national laboratories, and utilities. Experts from the following organizations participated in the interviews: Electranix, Sandia National Laboratories, EPC Power, Siemens PTI, Siemens Gamesa, Southern Company, Quanta, ETAP, GE, Hickory Ledge, RTE France, Dominion Energy, and Hawaiian Electric. Of the 18 total expert interviews 5 were conducted only within Siemens with existing non-disclosure agreements (NDAs). Care has been taken to exclude information that said organizations are not willing to publicly share. The following section provides a summary of the key takeaways from the interviews.

1. *What is the impact of inverter-dominated systems on system protection schemes like overcurrent, directional, and distance protection?*

Interviewed experts agree that IBR dominated systems can have significant impacts on system protection schemes. In particular, the GFM inverter vendors have proprietary controls that can lead to different behavior during and after faults. The consensus was that GFM inverters should provide enough negative sequence current for protection schemes (particularly distance/directional/overcurrent protection) to work appropriately. The phase selection logic for distance protection can be severely impacted and directional/distance protection schemes that rely on overcurrent supervision can also be affected by low overcurrent's from IBRs. Moreover, GFM inverters are known to have current-limiting strategies to protect the inverter hardware, and the way in which current is limited can impact protection. In particular, non-sinusoidal currents can impact RMS-based protection schemes. To overcome the above challenges, one utility are using Directional Comparison Blocking (DCB) protection, all-fiber differential, or continuing use of POTT schemes. Moreover, some utilities provide interconnection guidance for transmission systems that require high-side Y transformers for sourcing zero sequence currents during faults. Since most faults are line-to-ground faults – fault current does not drop significantly from relay perspective because of ground path from transformer.

2. *Can you give examples of protection mis-operation due to inverters from field? What was the cause of mis-operation? Did you take any steps to address the mis-operation?*

Most protection mis-operations are reported and presented by utilities at PSRC and similar conferences but focus on GFL inverters. Experts expect that there will be real-world challenges with GFM IBR-dominated systems:

- 1) how to supply negative sequence current from IBRs for over-current and distance protection to work correctly; and
- 2) how to improve phase-selection logic.

Experts shared that negative sequence was not required traditionally by GFL IBRs, but new standards and grid codes require it. Similar requirements are expected for GFM IBRs. Experts also agreed that phase-to-phase faults can lead to mis-operation (particularly of directional protection relays).

Some utilities have interconnection agreements that requires delta-Y transformer connections that can provide zero sequence current during faults – this is a fairly common practice and has been echoed

by other experts we interviewed. Since a majority of faults are single-line-to ground the zero-sequence current contribution from the transformers mitigates most protection mis-operations. Some utilities with high solar penetration have shared that faults typically occur during storms at night – during such events IBRs trip before system protection does.

Consultants doing simulation studies have also shared that they see issues with protection behavior far more frequently than in the past and utilities are recording more and more events for FRT studies – one such example is the well-known event reported by NERC on the Odessa, TX event in 2021. One expert also suggested that their blackstart tests and protection studies have shown that under-voltage protection would not work properly.

3. Have you seen examples of issues after the fault is cleared, such as the inverters going unstable, not resynchronizing, or causing other false trips?

Field examples in general were limited as there are few, if any, GFM IBR dominated systems in practice. In one expert interview with a consultant the feedback was that generally simulations are good but Functional Acceptance Testing (FAT - quality check for customer) shows issues with voltage recovery after the fault is cleared. Inverters do a ramp after fault is cleared (similar to a blackstart) which is different from the behavior of synchronous generators and may cause issues. In one interview, it was reported that oscillations have been observed after a fault was cleared in the field – esp. after inverters came out of current-limiting mode. Inverters can also trip as consequence of the oscillations during the post-fault recovery. It was not clear if oscillations were a particular situation for the specific system under study and IBR vendor. Utilities have reported problems with momentary cessation of IBRs – an issue also documented in several NERC reports.

4. Should the inverters try to behave more like synchronous machines to adapt to existing protection schemes, or should the protection system adapt to inverter-dominated systems?

Although this question is more philosophical, experts tend to agree that a combination of both would be ideal, i.e., change is required on both sides. Experts concur that complete protection philosophy cannot be changed yet the advantages of fast-responding IBRs should not be compromised simply to satisfy existing protection schemes that have been developed over decades for synchronous machine dominated systems. The challenge is that utilities need to maintain reliability and if IBRs behave more like synchronous machines then there is better system integrity. For this reason, some experts prefer higher fault currents and sinusoidal voltages, however, based on the GFM control schemes and vendor this is not always possible. To avoid hardware overdesign, it will be imperative to co-design inverter controls and system protection schemes. Experts tend to agree that GFM IBRs should still be required to provide, e.g., 1.5 p.u. current over 15-30 cycles during faults and protection system should adjust to these lower overcurrent limitations. To aid existing protection schemes, IBRs should also provide predictable negative sequence currents. Some experts argued that fault-currents in IBR dominated systems may be similar to synchronous machine dominated systems as not all GFM IBRs will be operating at their limit. Moreover, it was clear that several HVDC connected wind farms are being commissioned and protection for such offshore wind farms can be properly designed. Protection for GFM-IBRs dominated meshed AC systems is more challenging.

To overcome challenges in IBR-dominated systems, some experts advocate for adaptive protection schemes, yet such schemes require communication overhead unless they can be switched in an offline fashion and are therefore not preferred by some utilities. The preferred solution from at least one utility was to have one protection setting that covers all operating conditions.

5. *Does load encroachment in 100% IBR grids cause distance relays to trip? It might be that distance relays see mis-operations as fault since the fault currents will be much less in IBRs.*

The response from experts was mixed and inconclusive – some see load encroachment as a nuisance, while others don't see it affecting their distance protection schemes. Some utilities do not allow GFL IBRs to provide any reactive power (this may not be realistic in the future). It was argued that load encroachment leads to false trips only if IBRs provide reactive power. In fact, under such circumstances, the fault sometimes is not even within the zone of the distance relay and can lead to mis-operation. Other utilities have studies underway but do not expect issues – the underlying reason being that distance protection is immune to current and simply measuring the resistance.

Other expert interviews paint a slightly different picture – those experts argue that we will likely see more and more false trips because surge currents of large induction machines and transformers will create high “normal” load currents. In such settings, load encroachment may block relays due to low fault current contribution. Some have studied capacitor switching for line voltage regulation and have not yet seen issues with protection, but this could also be a major concern soon. CIGRE has recently published a technical report about the effect of load types on IBR behavior / protection system that provides further details [95].

6. *What is the effect of the grid-forming dynamic response on out-of-step protection tripping and power swing blocking protection at the transmission level?*

Industry experts affirmed that IBRs can vary their power output very rapidly (either on command or due to natural variability) and can also jump phase quickly, therefore, protection settings used for synchronous machines may not work properly for IBR dominated systems and will need to be tuned to prevent mis-operations. Of the four interviewed utilities, none seem to use out-of-step protection or power swing blocking except for generator protection. Nonetheless, the feedback we got was that utilities are concerned about power swings. One consulting firm we interviewed has already seen an issue with power swings in field – in particular, they observed that a battery energy storage system (BESS) IBR responded aggressively to frequency signals by ramping up/down active power output quickly. The BESS IBR was so fast that power swing relays detected the high active power rise and opened the line leading to a false trip. Another example provided was for a system in Australia that was tested against black-start and islanding events (in simulation) where out-of-step protection led to islanding of the system because the number of synchronous generators has been reduced.

7. *What analytic and simulation models are needed for grid-forming inverters under fault scenarios that will aid protection engineers in conducting accurate short-circuit studies?*

Industry experts agree that using an EMT simulation software such as PSCAD is necessary for analyzing behavior of GFM inverters under fault scenarios. However, several experts argued that running EMT simulation is time consuming and is not a feasible solution for large scale GFM IBR penetration and better tools are needed. To properly determine and update protection settings, industry experts would like to see accurate GFM inverter models (particularly FRT settings and behavior during over-current limiting operations) in phasor-based protection tools like Aspen and PSS/CAPE. Similarly, experts would also like better capabilities to model commercial protection relays in EMT simulation tools such as PSCAD. Although generic relay models are available in PSCAD, an extensive commercial relay library available through protection software such as PSS/CAPE is needed. Industry experience with co-simulation platforms is mixed, while some see potential benefits at least in the interim, i.e., until protection tools can incorporate detailed GFM inverter models or EMT tools are able to accommodate extensive relay libraries, others argue that co-

simulation does not yield too many benefits and are too advanced for many protection engineers. Protection engineers are more familiar with using simple, phasor-based fault-study tools, so GFM performance should be standardized during faults to make studies with generic models feasible. While this simplistic approach will reduce the need to have detailed black-box models from vendors and may work for over-current protection, it is unclear whether it would work also for distance protection schemes. Several vendors have the required interfaces to conduct co-simulation with PSCAD, however, none of them are currently using this functionality. As the industry currently stands, simulations are mostly done in one or two tools that combine transient stability analysis with protection modeling, such as PSS/E+PSS/CAPE and ETAP.

4. SUMMARY AND CONCLUSION

Experts agree that GFM IBR dominated systems will lead to a significant impact on system protection schemes, particularly distance, directional, and power swing schemes. The severity of the impact depends on the GFM and GFL IBR controls. Post-fault recovery, oscillations, or protection mis-operations have also been identified as major concerns.

Expert interviews and the literature review also confirm that a big gap exists between where the literature is today and what is needed for protecting a 100% IBR power system. There is hardly any literature or knowledge on

- Measurements of fault currents generated by GFM IBRs (with very few exceptions mentioned in Section 2.3),
- FRT strategies for GFM IBRs with a focus on the correct operation of protection relays from a system protection perspective,
- Protection relay functions that respond correctly to fault currents measured in IBR-dominated systems, also during different operation models of GFM IBRs such as black start,
- Interaction between GFM and GFL IBRs during and after faults and their impact on the protection system,
- Accurate models for GFM IBRs and protection relays for system protection studies,
- Tools for protection studies in IBR-dominated systems, and
- Input for future protection standards.

The existing literature and studies are focused on related but different areas such as

- Several publications and studies have investigated system protection for GFL IBRs but not for GFMs.
- Several publications have developed FRT strategies for GFM IBRs with different objectives, such as voltage balancing, transient stability and synchronization improvements, meeting grid codes, etc., but not from a system protection perspective.
- Publications and studies on protection for inverter-dominated systems are limited to microgrids and offshore wind farms radially connected to an HVDC terminal.

The experts agree that there is extensive R&D needed to fully comprehend the effect of GFM IBRs with different control structures and current limiting techniques for different protection functions such as distance, directional, and differential during and after faults. This will then help in developing relay algorithms and protection schemes compatible with these sources.

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